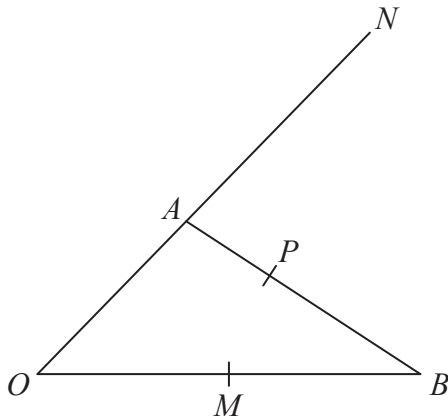




OAN , OMB and APB are straight lines.

$$AN = 2OA.$$

M is the midpoint of OB .

$$\overrightarrow{OA} = \mathbf{a} \quad \overrightarrow{OB} = \mathbf{b}$$

$$\overrightarrow{AP} = k\overrightarrow{AB} \text{ where } k \text{ is a scalar quantity.}$$

Given that MPN is a straight line, find the value of k .